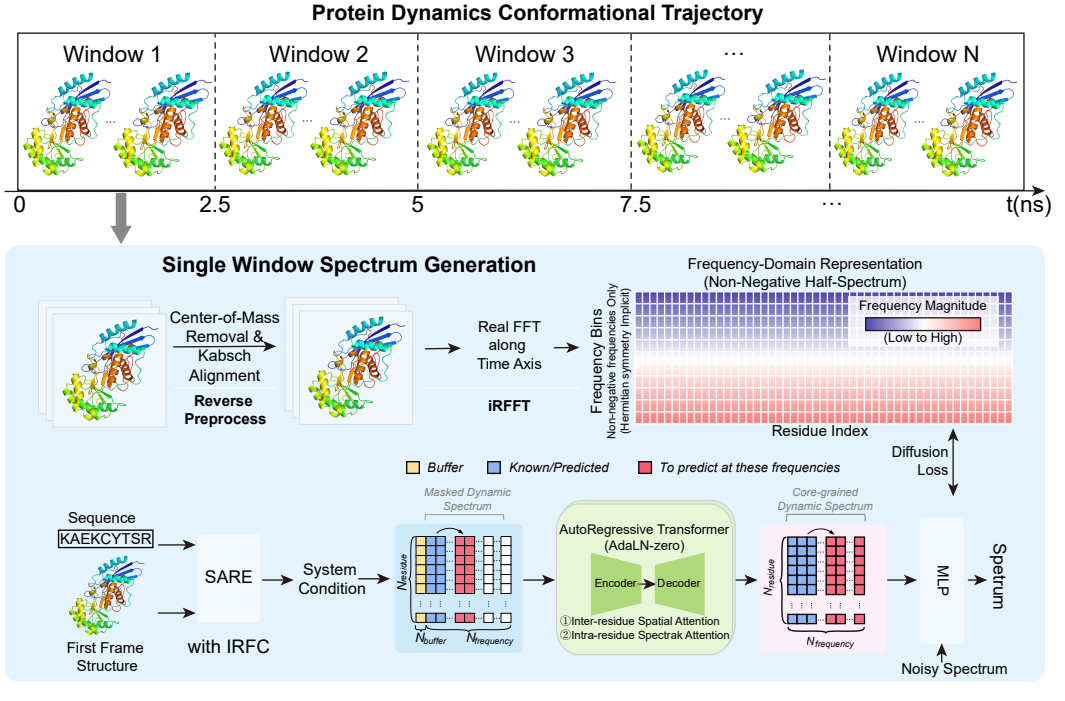
Protein Dynamics Conformational Trajectory
Window 1
Window 2
Window 3
…
Window N
0
2.5
5
7.5
…
t(ns)
Single Window Spectrum Generation
Center-of-Mass Removal & Kabsch Alignment
Reverse Preprocess
Real FFT along Time Axis
iRFFT
Frequency-Domain Representation (Non-Negative Half-Spectrum)
Frequency Bins
Non-negative frequencies Only (Hermitian symmetry Implicit)
Frequency Magnitude
(Low to High)
Residue Index
Diffusion Loss
Buffer
Known/Predicted
To predict at these frequencies
Sequence
KAEKCYTSR
First Frame Structure
SARE
with IRFC
System Condition
Masked Dynamic Spectrum
AutoRegressive Transformer (AdaLN-zero)
Encoder
Decoder
①Inter-residue Spatial Attention
②Intra-residue Spectrak Attention
Core-grained Dynamic Spectrum
MLP
Spetrum
Noisy Spectrum